

An Artin group with no geometric $CAT(0)$ action

An edited version with a 29-generator counterexample

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Abstract

We construct an Artin group on 29 generators that admits no proper, cocompact isometric action on a nonempty proper $CAT(0)$ space. This refutes the $CAT(0)$ conjecture for Artin groups.

About this version. This is an edited version of the OpenAI preprint *An Artin group with no geometric $CAT(0)$ action* (September 23, 2026), which gives a counterexample on 116 generators. The construction and all arguments other than the one noted below are OpenAI's; the original is distributed under the Apache License 2.0 at <https://github.com/openai/math>.

What changed. The proof of Proposition 3.1 is replaced by a shorter argument that needs braid blocks of 11 strands instead of 40. With the construction of Section 7 otherwise unchanged, the counterexample shrinks from 116 to 29 generators, and the generator counts have been updated to match.

Verification. The repository's Lean formalization of the full theorem was adapted to the 29-generator matrix and checks it using only the axioms `propext`, `Classical.choice` and `Quot.sound`. This version was prepared with the assistance of Claude (Anthropic).

Contents

1	Introduction	2
2	Stable lengths and centralizer characters	4
3	A quantitative constraint from braid groups	7
4	Six asymptotic sectors	10
5	Labelled triangular diagrams	15
6	A local winding count	21
7	The finite Artin matrix and the final contradiction	23

1 Introduction

Let S be finite and let $M = (m_{st})_{s,t \in S}$ be symmetric, with $m_{ss} = 1$ and

$$m_{st} \in \{2, 3, \dots\} \cup \{\infty\} \quad (s \neq t).$$

The Artin group A_M has generators σ_s , $s \in S$. For each finite m_{st} , its defining relation equates the two alternating words of length m_{st} starting with σ_s and σ_t . A label ∞ imposes no relation.

Throughout, an action is called *geometric* if it is an isometric action on a nonempty proper CAT(0) metric space X , is proper, and is cocompact. Here a metric space is proper when its closed bounded subsets are compact; action properness means that

$$\{g : gK \cap K \neq \emptyset\}$$

is finite for every compact $K \subset X$; and cocompactness means that $GK = X$ for some compact K . Our CAT(0) convention is the geodesic triangle-comparison inequality with Euclidean comparison triangles. The explicit properness assumption on X also follows from the other action hypotheses for a length space [5, Exercise I.8.4(1)].

Charney asked which Artin groups are CAT(0) [6, Problem 4]. The affirmative expectation that every finite-rank Artin group admits a geometric action is recorded by Haettel as Conjecture 1.1 of [12]. A geometric action would make the group's large-scale geometry accessible through geodesic convexity and Euclidean comparison. The challenge is compatibility: constructing actions for the rank-two braid subgroups does not by itself supply an action of the whole group.

There are substantial positive results. Right-angled Artin groups, whose finite off-diagonal labels are all two, have finite nonpositively curved cubical classifying spaces and hence geometric CAT(0) actions [7, preprint Theorem 3.1.1]. Brady and McCammond constructed nonpositively curved two-dimensional complexes for the three-generator Artin groups with all three finite labels at least three [2, preprint Theorem 4]. Haettel proved the conjecture for XXL type, where every finite off-diagonal label is at least five [12, Theorem 1.4]. For classical braid groups, Brady and McCammond established the five-strand case [3, Theorem B], Haettel, Kielak, and Schwer the six-strand case [13, preprint Corollary 4.18], and Jeong the seven-strand case [14, preprint Corollary 14].

In the opposite direction, Brady and Crisp found Artin groups with two-dimensional classifying spaces for which the least dimension of a geometric CAT(0) action is three [1, preprint Theorems B and 3.1]. Their examples show that an obstruction in one dimension can disappear in a higher-dimensional CAT(0) space. To refute the conjecture, an obstruction must instead survive every dimension.

Theorem 1.1. *There is an Artin matrix M on 29 generators, with every off-diagonal label in $\{2, 3, \infty\}$, such that A_M admits no geometric action. One explicit matrix is given in Definition 7.1.*

Thus Theorem 1.1 gives a negative resolution of the CAT(0) conjecture for Artin groups. In particular, the conclusion excludes every nonempty proper CAT(0) space, regardless of dimension or the choice of a complex or metric.

The construction and the obstruction

The small group that detects the obstruction is

$$T = \langle a, b, c \mid aba = bab, bcb = cbc, cac = aca \rangle.$$

To each of its three cyclic braid pairs we attach an 11-strand braid group, sharing the first two standard generators of that block with the pair. Each attachment adds 8 generators. Two further generators d, e , each commuting with a, b, c and having no other finite incident labels, make T their simultaneous centralizer; Section 7 proves this equality. This gives $3 + 3 \cdot 8 + 2 = 29$ generators.

For an isometry coming from a geometric action, write $\ell(g)$ for its stable translation length, the limit of $d(x, g^n x)/n$. Section 2 turns these lengths on commuting elements into Euclidean pairings and proves that a pairing with a fixed element is additive on its whole centralizer. The calculation in Section 3 uses these standard geometric properties to constrain the braid relations, without requiring the braid blocks themselves to act cocompactly. A long block forces

$$\frac{\ell((uv)^3)}{3\ell(u^2)} = \frac{\ell((uv)^3)}{3\ell(v^2)} > \frac{1}{2}.$$

If these inequalities held on all three pairs in a geometric action of T , the commuting pairs

$$(a^2, (ab)^3), (b^2, (ab)^3), (b^2, (bc)^3), (c^2, (bc)^3), (c^2, (ca)^3), (a^2, (ca)^3)$$

would give six asymptotically Euclidean sectors, each of angle less than $\pi/3$. Their cyclic union would have total angle less than 2π .

A related angle-sum obstruction appears in Crisp's study of two-dimensional Bestvina–Brady groups [8, Lemma 1.3 and the following remark]. His argument assumes covering dimension two and a proper action in which each isometry attains its minimum displacement.

The main geometric step is to turn this short angular circuit into a contradiction that remains valid in arbitrary dimension. Coordinates coming from an affine mirror action separate disjoint regions of the sector disk at large scale. This action is the equilateral-triangle specialization of Digne's reflection-coordinate homomorphism [10, Proposition 2.1]. CAT(0) geodesics provide a second disk whose image stays far from the cone point. We transfer the resulting sphere to a triangular presentation complex, using a bound valid at every point of every parameter triangle. A local winding count then detects a target triangle with nonzero signed multiplicity, whereas a direct folding argument makes every such multiplicity zero.

The triangular presentation and its girth-six link are the $m = n = p = 3$ specialization of Brady and McCammond's construction [2, preprint Lemmas 2–3]. We prove the facts needed here, including the signed-count and controlled-approximation statements. The latter comparison between local degree and global signed counts is formulated entirely in terms of distances and finite diagrams. It does not require a triangulation or a dimension bound on the unknown CAT(0) space.

Finally, the simultaneous-centralizer construction makes the action on T geometric: a suitable intersection of two displacement sublevel sets is closed and convex, and T acts cocompactly on it. Its stable lengths are the ambient stable lengths. The braid inequalities and the six-sector obstruction therefore apply to the same action.

Organization and notation

Sections 2 and 3 establish the length calculation. Sections 4–6 prove the obstruction for T . Section 7 gives the matrix, the algebraic embeddings, and the centralizer argument completing Theorem 1.1. The substantial geometric, algebraic, and diagrammatic ingredients are proved below. We use elementary finite planar topology for regular neighbourhoods of graphs and winding numbers; no Artin-group asphericity theorem is an input.

We use B_l for the braid presentation on $s_1, \dots, s_{l-1}$: adjacent generators satisfy the length-three braid relation, and generators at distance at least two commute. For commuting elements g, h , the notation $\langle g, h \rangle$ will denote the polarization of their squared stable lengths, defined in Lemma 2.3.

2 Stable lengths and centralizer characters

Our objective is a pairing of stable lengths that is additive on an entire centralizer, even when its elements do not commute with one another. The geometric facts used here are standard: attainment of minimum displacement under a geometric action and the structure of its minimizing set are treated in Bridson–Haefliger [5, II.6.8(5) and II.6.10(2)]; the Euclidean structure on commuting translations is central to the Flat Torus Theorem [5, II.7.1]. We derive precisely the length identities needed for the braid calculation directly from triangle comparison, without using a splitting theorem.

Lemma 2.1. *In a $CAT(0)$ space, geodesic segments are unique and depend continuously on their endpoints. If $\alpha, \beta : [0, 1] \rightarrow X$ are geodesics parametrized proportionally to length, then*

$$d(\alpha(t), \beta(t)) \leq (1 - t)d(\alpha(0), \beta(0)) + td(\alpha(1), \beta(1)). \quad (2.1)$$

If m is the midpoint of $[v, w]$, then

$$d(z, m)^2 \leq \frac{d(z, v)^2 + d(z, w)^2}{2} - \frac{d(v, w)^2}{4}. \quad (2.2)$$

For any four points p, q, r, s ,

$$d(p, r)^2 + d(q, s)^2 \leq d(p, q)^2 + d(q, r)^2 + d(r, s)^2 + d(s, p)^2. \quad (2.3)$$

Proof. For uniqueness, split one of two segments with the same endpoints at an arbitrary point and use the other segment as the third side of the resulting triangle. Its comparison triangle is a straight segment, so comparison identifies the two points with the same proportional parameter. For two segments with a common initial point, comparison bounds their distance at parameter t by t times the distance of their terminal points. The analogous statement holds for a common terminal point. Inserting the segment from $\alpha(0)$ to $\beta(1)$ between α and β , and using the triangle inequality, proves (2.1). This bound also proves continuous dependence on the endpoints; dependence on the parameter follows from constant-speed parametrization.

The Euclidean identity for a median, together with comparison, gives (2.2). Let m now be the midpoint of $[p, r]$. The triangle inequality and $(A + B)^2 \leq 2A^2 + 2B^2$ give

$$d(q, s)^2 \leq 2d(q, m)^2 + 2d(s, m)^2.$$

Apply (2.2) to each term on the right to obtain (2.3). □

For an isometry g and a base point o , define its *stable length* by

$$\ell(g) = \lim_{n \rightarrow \infty} \frac{d(o, g^n o)}{n}. \quad (2.4)$$

Here the limit exists: the sequence $a_n = d(o, g^n o)$ is subadditive, and, for fixed k and $n = qk + r$ with $0 \leq r < k$, $a_n \leq qa_k + a_r$. Consequently $\limsup a_n/n \leq a_k/k$ for every k , and the limit equals $\inf_{k \geq 1} a_k/k$. Changing o changes a_n by at most twice the distance between the two base points. Thus the length is independent of o , is invariant under conjugation, and satisfies

$$\ell(g^m) = |m|\ell(g) \quad (m \in \mathbb{Z}). \quad (2.5)$$

For negative m , use $d(o, g^{-n}o) = d(o, g^n o)$; for positive m , use the subsequence indexed by multiples of m in (2.4).

Lemma 2.2. *Suppose G acts geometrically on X . Every $g \in G$ attains its minimum displacement, and*

$$\ell(g) = \min_{x \in X} d(x, gx).$$

If this number is positive, g preserves a geodesic line and translates along it by $\ell(g)$. Moreover, $\ell(g) = 0$ if and only if g has finite order.

Proof. Set $D = \inf_x d(x, gx)$, and choose x_n with $d(x_n, gx_n) \rightarrow D$. Write $x_n = h_n p_n$ with p_n in a compact set K whose G -translates cover X . After discarding finitely many terms, $q_n = h_n^{-1} g h_n$ moves p_n by at most $D + 1$. The closed $(D + 1)$ -neighborhood K' of K is closed and bounded, hence compact. Both p_n and $q_n p_n$ belong to K' , so $q_n K' \cap K' \neq \emptyset$. Action properness gives only finitely many possibilities for q_n . Pass to a subsequence on which $q_n = q$ and $p_n \rightarrow p \in K$. Then $d(p, qp) = D$. A fixed one of the conjugators h_n transfers p to a minimizer for g .

Let x be such a minimizer. If $D > 0$, let m be the midpoint of $[x, gx]$; then gm is the midpoint of $[gx, g^2x]$. Comparison in the triangle with vertices x, gx, g^2x gives

$$D \leq d(m, gm) \leq \frac{1}{2}d(x, g^2x) \leq D.$$

Thus $d(x, g^2x) = 2D$. We verify that all consecutive orbit segments form a line. Suppose the concatenation from x to $g^i x$ is geodesic. In the triangle with vertices $x, g^i x, g^{i+1} x$, the points at distance D from $g^i x$ along the two incident sides are $g^{i-1} x$ and $g^{i+1} x$. Their distance is $2D$, by the preceding equality and translation by g^{i-1} . Their Euclidean comparison points can be at distance $2D$ only if the comparison angle at $g^i x$ is π . Hence $d(x, g^{i+1} x) = (i + 1)D$. Induction, followed by translation by all powers of g , proves that the segments $[g^i x, g^{i+1} x]$, $i \in \mathbb{Z}$, form an isometrically parametrized line. In particular $\ell(g) = D$.

If $D = 0$, the minimizer is fixed, so $\ell(g) = 0$. Its stabilizer is finite by action properness applied to the compact singleton $\{x\}$; thus g has finite order. Conversely, finite order immediately gives zero stable length. $\square$

For commuting isometries of a complete CAT(0) space, Rodenhausen [15, Theorem 14.6] proves a parallelogram identity for infimal displacements. The next proof works directly with stable lengths.

Lemma 2.3. *For pairwise commuting isometries $g_1, \dots, g_k$ of a CAT(0) space, the function*

$$v \mapsto \ell(g^v), \quad g^v = g_1^{v_1} \cdots g_k^{v_k}, \quad v \in \mathbb{Z}^k,$$

extends uniquely to a continuous seminorm on $\mathbb{R}^k$ whose square is a positive semidefinite quadratic form. For commuting g, h , put

$$\langle g, h \rangle = \frac{\ell(gh)^2 - \ell(g)^2 - \ell(h)^2}{2}. \quad (2.6)$$

These pairings are the bilinear form on every commuting family; in particular, $\langle g^i, h^j \rangle = ij \langle g, h \rangle$. Lengths and pairings are invariant under simultaneous conjugation.

Proof. Write $\nu(v) = \ell(g^v)$ for integer v . Commutation, the triangle inequality, and (2.5) give

$$\nu(u + v) \leq \nu(u) + \nu(v), \quad \nu(mv) = |m|\nu(v),$$

$$|\nu(u) - \nu(v)| \leq \sum_{i=1}^k |u_i - v_i| \ell(g_i).$$

Define $\nu(v/q) = \nu(v)/q$ for integers $q > 0$. Homogeneity makes this well-defined on $\mathbb{Q}^k$, and the displayed Lipschitz bound gives a unique continuous extension to $\mathbb{R}^k$. It is a seminorm.

Apply (2.3) to the four points

$$o, \quad g^{nu}o, \quad g^{n(u+v)}o, \quad g^{nv}o$$

in that cyclic order. Divide by n^2 and pass to the limit to obtain, with $Q = \nu^2$,

$$Q(u+v) + Q(u-v) \leq 2Q(u) + 2Q(v).$$

This initially holds for integer vectors and extends by scaling and continuity to real vectors. Apply it to $u+v, u-v$ and use $Q(2u) = 4Q(u)$ to obtain the opposite inequality. Hence

$$Q(u+v) + Q(u-v) = 2Q(u) + 2Q(v). \quad (2.7)$$

For completeness, polarization here requires only elementary algebra. Set $B(u, v) = (Q(u+v) - Q(u-v))/4$. It is symmetric and continuous, and (2.7) gives

$$B(u+w, v) + B(u-w, v) = 2B(u, v).$$

Taking $w = u$ gives $B(2u, v) = 2B(u, v)$, since $B(0, v) = 0$. Next take $u = (a+b)/2, w = (a-b)/2$ to obtain additivity in the first variable. Integer and rational homogeneity follow, and continuity gives real homogeneity. Symmetry gives the same properties in the second variable. Finally $B(u, u) = Q(u) \geq 0$, so B is positive semidefinite; expanding $Q(u+v)$ recovers (2.6). Conjugation invariance follows from that of stable length. $\square$

The following strengthening of bilinearity is crucial. The elements h_1, h_2 in its statement need not commute with one another. Real translation characters also underlie Bridson's obstruction to proper semisimple actions on two-dimensional CAT(0) spaces [4, Proposition 3.1].

Lemma 2.4. *For a geometric action and a fixed $g \in G$, the function*

$$C_G(g) \longrightarrow \mathbb{R}, \quad h \longmapsto \langle g, h \rangle$$

is a group homomorphism. Thus $\langle g, h_1 h_2 \rangle = \langle g, h_1 \rangle + \langle g, h_2 \rangle$ whenever both h_1 and h_2 commute with g .

Proof. If $\ell(g) = 0$, positive semidefiniteness in Lemma 2.3 implies $\langle g, h \rangle = 0$ for every $h \in C_G(g)$: the nonnegative quadratic $\ell(g^t h)^2 = 2t\langle g, h \rangle + \ell(h)^2$, for integer t of both signs, forces the coefficient to vanish.

Now let $D = \ell(g) > 0$ and choose a minimizer x . By Lemma 2.2, $d(x, g^n x) = nD$. Define

$$b(z) = \lim_{n \rightarrow \infty} (d(z, g^n x) - nD).$$

The terms decrease by the triangle inequality and are bounded below by $-d(z, x)$, so the limit exists. The function b is 1-Lipschitz.

Fix $h \in C_G(g)$ and $z, w \in X$. Put $p_n = g^n x$ and $q_n = h^{-1} g^n x$. Since h commutes with g , $d(p_n, q_n) = d(x, h^{-1} x)$ is constant. Two orderings in (2.3) give

$$|d(z, p_n)^2 + d(w, q_n)^2 - d(w, p_n)^2 - d(z, q_n)^2| \leq d(z, w)^2 + d(p_n, q_n)^2.$$

Every long distance on the left is $nD + O(1)$. More precisely, $d(z, p_n) = nD + b(z) + o(1)$ and $d(z, q_n) = d(hz, p_n) = nD + b(hz) + o(1)$, and similarly for w . Dividing the inequality by $2nD$ and passing to the limit therefore gives

$$b(hz) - b(hw) = b(z) - b(w). \quad (2.8)$$

Consequently $\chi(h) = b(x) - b(hx)$ is a homomorphism on $C_G(g)$: apply (2.8) with $z = kx, w = x$ to compute $\chi(hk) = \chi(h) + \chi(k)$. Also $b(x) = 0$, $b(gx) = -D$, and hence $\chi(g) = D$. Lipschitz continuity, applied to h^n , gives

$$n|\chi(h)| = |b(x) - b(h^n x)| \leq d(x, h^n x), \quad \text{so } |\chi(h)| \leq \ell(h).$$

Apply this last inequality to $g^t h$ for every integer t . By Lemma 2.3,

$$(tD + \chi(h))^2 \leq t^2 D^2 + 2t\langle g, h \rangle + \ell(h)^2.$$

After cancelling the quadratic terms, the inequality for arbitrarily large positive and negative t forces $D\chi(h) = \langle g, h \rangle$. Thus the pairing is a multiple of the homomorphism χ , as asserted. $\square$

3 A quantitative constraint from braid groups

Let B_l denote the group on $s_1, \dots, s_{l-1}$ with relations

$$s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}, \quad s_i s_j = s_j s_i \quad \text{if } |i - j| \geq 2.$$

Throughout this section the same notation may denote elements of another group satisfying these relations. No injectivity or cocompactness of the resulting braid subgroup is needed for the length estimates.

Proposition 3.1. *Suppose a group acting geometrically on a $CAT(0)$ space contains elements $s_1, \dots, s_{10}$ satisfying the B_{11} relations, with $\ell(s_1^2) > 0$. Then*

$$\left(\frac{\ell((s_1 s_2)^3)}{3\ell(s_1^2)} \right)^2 \geq \frac{7}{27} > \frac{1}{4}, \quad (3.1)$$

and in particular

$$\frac{\ell((s_1 s_2)^3)}{3\ell(s_1^2)} = \frac{\ell((s_1 s_2)^3)}{3\ell(s_2^2)} > \frac{1}{2}.$$

For B_4 , Crisp and Paoluzzi compute translation components along an axis of a central element under a minimal geometric action [9, Proposition 5]; minimality means that no proper nonempty closed convex subset is invariant.

To prove Proposition 3.1, we compare central elements of the initial braid subblocks using the centralizer characters of Section 2. For a general B_l with $l \geq 3$, define

$$L_1 = 1, \quad L_j = s_{j-1} L_{j-1} s_{j-1}, \quad Z_r = L_1 \cdots L_r, \quad N_r = \binom{r}{2}. \quad (3.2)$$

Here $2 \leq j \leq l$ and $1 \leq r \leq l$.

Lemma 3.2. *The element L_j commutes with s_i for $i \leq j - 2$, and Z_r commutes with every s_i for $i < r$. In particular,*

$$Z_2 = s_1^2, \quad Z_3 = (s_1 s_2)^3.$$

The element $\delta = s_1 \cdots s_{l-1}$ satisfies

$$\delta s_i \delta^{-1} = s_{i+1} \quad (1 \leq i \leq l-2). \quad (3.3)$$

In any geometric action of the ambient group,

$$\langle s_i^2, Z_r \rangle = \frac{\ell(Z_r)^2}{N_r} \quad (2 \leq r \leq l, \ 1 \leq i < r). \quad (3.4)$$

Proof. Induct on j for the assertion about L_j . For $i \leq j - 3$, both s_{j-1} and L_{j-1} commute with s_i . For $i = j - 2$, put $a = s_{j-2}$, $b = s_{j-1}$, and $C = L_{j-2}$. Then $bC = Cb$ and $L_j = baCab$, so the braid relation gives

$$aL_j = abaCab = babCab = baCbab = baCaba = L_j a.$$

This includes the case $j = 3$, where $C = L_1 = 1$; the case $j = 2$ is empty. In particular, L_j commutes with each earlier L_i , because the word for L_i uses only $s_1, \dots, s_{i-1}$.

Next induct on r for Z_r . Both Z_{r-1} and L_r commute with s_i for $i \leq r - 2$. The only remaining generator is $s = s_{r-1}$. It commutes with Z_{r-2} . With $L = L_{r-1}$ we have $L_r = sLs$, and the already proved commutation of L with L_r says $LsLs = sLsL$. Therefore

$$s(LL_r) = sLsLs = LsLs^2 = (LL_r)s,$$

which proves the assertion. The case $r = 2$ is immediate. Finally

$$Z_3 = s_1^2 s_2 s_1^2 s_2 = s_1 (s_1 s_2 s_1) s_1 s_2 = s_1 (s_2 s_1 s_2) s_1 s_2 = (s_1 s_2)^3.$$

To verify (3.3), in δs_i commute the last s_i left past $s_{i+2}, \dots, s_{l-1}$, replace $s_i s_{i+1} s_i$ by $s_{i+1} s_i s_{i+1}$, and commute the new s_{i+1} left past $s_1, \dots, s_{i-1}$. The result is $s_{i+1} \delta$.

For the length identity, apply the character $h \mapsto \langle Z_r, h \rangle$ from Lemma 2.4. All s_i , $i < r$, are in its domain. Applying it to each adjacent braid relation shows that their values are equal; call the common value α . The positive word for L_j has $2(j-1)$ letters, so that for Z_r it has $2N_r$ letters. Thus

$$\ell(Z_r)^2 = \langle Z_r, Z_r \rangle = 2N_r \alpha, \quad \langle s_i^2, Z_r \rangle = 2\alpha,$$

which proves (3.4), including when $\ell(Z_r) = 0$. $\square$

Proof of Proposition 3.1. We work in B_l with $l \geq 4$ and specialize to $l = 11$ only at the end. Put $g = s_1^2$ and $D = \ell(g) > 0$. Since $s_2 = (s_1 s_2) s_1 (s_1 s_2)^{-1}$, also $\ell(s_2^2) = D$. For $2 \leq r \leq l$ put

$$Q_r = \frac{\langle g, Z_r \rangle}{D^2}, \quad \text{so that} \quad \ell(Z_r)^2 = N_r Q_r D^2$$

by (3.4) with $i = 1$. Thus $Q_2 = 1$, and $Q_3 = \ell(Z_3)^2 / (3D^2) = 3R_3$, where $R_3 = \ell((s_1 s_2)^3)^2 / (9D^2)$ is the square of the ratio in (3.1).

A recursion for Q_r . All s_j with $j \geq 3$ commute with g . Applying the character of g (Lemma 2.4) to the braid relation between s_j and s_{j+1} , $3 \leq j \leq l-2$, shows that $\langle g, s_j \rangle$ is independent of $j \geq 3$; write it as $c_0 D^2 / 2$. Put $x = Q_3 - 1$; since $Z_3 = g L_3$, we have $\langle g, L_3 \rangle = x D^2$. For $j > 3$ the recursive definition gives

$$L_j = (s_{j-1} \cdots s_3) L_3 (s_3 \cdots s_{j-1}),$$

and every displayed factor lies in $C_G(g)$ (for L_3 by Lemma 3.2). Keeping L_3 as one factor, the character gives $\langle g, L_j \rangle = (x + (j-3)c_0)D^2$ for $j \geq 3$. Summing over the factors $L_2 = g, L_3, \dots, L_r$ of Z_r ,

$$Q_r = 1 + (r-2)x + \binom{r-2}{2}c_0 \quad (2 \leq r \leq l). \quad (3.5)$$

This calculation never expands the noncommuting s_2 -letters inside L_3 .

Two positivity constraints. First, $\ell(Z_l)^2 = N_l Q_l D^2 \geq 0$, so

$$Q_l = 1 + (l-2)x + \binom{l-2}{2}c_0 \geq 0. \quad (3.6)$$

Second, Z_{l-1} and Z_l commute, because every letter of the positive word for Z_{l-1} is some s_i with $i \leq l-2$. Applying the character of Z_l to that word, as in the proof of Lemma 3.2, each of its $2N_{l-1}$ letters contributes $\ell(Z_l)^2/(2N_l)$, so

$$\langle Z_{l-1}, Z_l \rangle = \frac{N_{l-1}}{N_l} \ell(Z_l)^2 = N_{l-1} Q_l D^2.$$

By Lemma 2.3, the element $W = Z_{l-1}^l Z_l^{-(l-2)}$ has

$$\begin{aligned} \frac{\ell(W)^2}{D^2} &= l^2 N_{l-1} Q_{l-1} - 2l(l-2)N_{l-1} Q_l + (l-2)^2 N_l Q_l \\ &= \frac{l(l-1)(l-2)}{2} (l Q_{l-1} - (l-2) Q_l), \end{aligned}$$

using $N_l = l(l-1)/2$ and $N_{l-1} = (l-1)(l-2)/2$. Hence $l Q_{l-1} - (l-2) Q_l \geq 0$, which by (3.5) reads

$$2 + (l-4)x - 2(l-3)c_0 \geq 0. \quad (3.7)$$

Elimination of c_0 . Add (3.6) to $(l-2)/4$ times (3.7). The coefficients of c_0 are $\binom{l-2}{2}$ and $-(l-2)(l-3)/2$, which cancel, leaving

$$\frac{l}{2} + \frac{l(l-2)}{4}x \geq 0, \quad \text{that is,} \quad x \geq -\frac{2}{l-2}.$$

Therefore

$$R_3 = \frac{1+x}{3} \geq \frac{l-4}{3(l-2)}, \quad (3.8)$$

which exceeds $1/4$ exactly when $l \geq 11$. For $l = 11$ it gives $R_3 \geq 7/27 > 1/4$. Finally $\ell(s_2^2) = \ell(s_1^2) = D$ gives the equality of the two displayed ratios. $\square$

Remark 3.3. The bound (3.8) is the best obtainable from (3.5), (3.6) and (3.7): at $x = -2/(l-2)$ and $c_0 = 2/((l-2)(l-3))$ one has $Q_{l-1} = Q_l = 0$. For $l = 10$ it gives only $R_3 \geq 1/4$, which is not enough for the strict sector inequality. Numerically, adding the positive semidefiniteness of the full Gram matrices of the commuting families built from disjoint and nested braid subblocks does not improve (3.8).

4 Six asymptotic sectors

Set

$$T = \langle a, b, c \mid aba = bab, bcb = cbc, cac = aca \rangle, \quad \mathcal{E} = \{(a, b), (b, c), (c, a)\},$$

and write $z_{uv} = (uv)^3$ for $(u, v) \in \mathcal{E}$.

Proposition 4.1. *There is no geometric action of T such that, for every $(u, v) \in \mathcal{E}$,*

$$\frac{\ell(z_{uv})}{3\ell(u^2)} = \frac{\ell(z_{uv})}{3\ell(v^2)} > \frac{1}{2}. \quad (4.1)$$

The denominators are positive in every geometric action: sending each of a, b, c to $1 \in \mathbb{Z}$ shows that their squares have infinite order, so Lemma 2.2 applies. The equality of the two ratios also follows from conjugacy of adjacent braid generators. We prove the proposition in Section 6. In this section, assume for contradiction that such an action on Y exists and fix $y \in Y$.

We will construct a parameter disk P made of six Euclidean sectors and maps $f_n : P \rightarrow Y$, indexed by positive integers n . Within each sector, image distances divided by n approach Euclidean distances. Mirror coordinates will ensure a second property: disjoint closed regions of P have images separated by a positive multiple of n . The total sector angle below 2π then lets us fill the same boundary by a second disk whose image stays a positive multiple of n from y . Sections 5 and 6 compare the resulting sphere with finite labelled diagrams to obtain the contradiction.

4.1 Orbit control and mirror coordinates

Let d_T be the word metric for a, b, c and their inverses. The following form of the Švarc–Milnor orbit comparison is standard; see [5, I.8.18–19]. Its direct proof records the constants used to turn mirror-coordinate separation into metric separation.

Lemma 4.2. *For a geometric action of T and any base point y , there are constants $R, C < \infty$ such that Ty is R -dense and*

$$d_T(s, t) \leq C(1 + d_Y(sy, ty)) \quad (s, t \in T). \quad (4.2)$$

Proof. If $TK = Y$ with K compact, take $R = \max_{p \in K} d(y, p)$. Every tp is within R of ty . For $L = d(sy, ty)$, divide the geodesic into $N = \max(1, \lceil L \rceil) \leq L + 1$ pieces, each of length at most one. Choose nearby orbit points at the division points, keeping the prescribed endpoint labels s, t . Each successive group increment q satisfies $d(y, qy) \leq 2R + 1$. There are finitely many such q : the compact ball $\overline{B}(y, 2R + 1)$ contains both y and qy , so properness applies. The largest word length of these increments, enlarged to at least one, can be used for C . Summing their word lengths proves (4.2). $\square$

Digne’s reflection-coordinate homomorphism sends an Artin generator associated to a reflection r to (e_r, r) in a semidirect product [10, Proposition 2.1]. We use its concrete version for the side reflections of an equilateral triangle and verify its formulas. The resulting coordinates will distinguish all six sectors.

Take an equilateral triangle in the Euclidean plane, and denote its three side lines by D_a, D_b, D_c . Let r_u be reflection in D_u , and let $\mathcal{D}$ be the set of all images of these three lines under the generated reflection group. Write

$$E = \bigoplus_{D \in \mathcal{D}} \mathbb{R}e_D, \quad \left\| \sum_D \xi_D e_D \right\|_1 = \sum_D |\xi_D|.$$

All vectors here have finite support. Each r_u preserves $\mathcal{D}$ and acts linearly on E by permuting its basis vectors.

Lemma 4.3. *The formulas*

$$u \cdot \xi = e_{D_u} + r_u \xi \quad (u = a, b, c)$$

define an affine isometric action of T on E . If $V(t) = t \cdot 0$, then

$$\|V(t) - V(s)\|_1 \leq d_T(s, t). \quad (4.3)$$

For each $(u, v) \in \mathcal{E}$, the elements u^2, z_{uv} act by pure translations with vectors

$$L_u = 2e_{D_u}, \quad L_{uv} = 2(e_{D_u} + e_{D_v} + e_{D_{uv}}), \quad (4.4)$$

where D_{uv} is the third mirror at $D_u \cap D_v$. The three lines D_{ab}, D_{bc}, D_{ca} are distinct from one another and from D_a, D_b, D_c .

Proof. Two incident side reflections generate the reflection group of three lines spaced at angles $\pi/3$ about their intersection. Their two length-three braid words have the same linear part. The translation part of uvu is

$$e_{D_u} + e_{r_u D_v} + e_{r_u r_v D_u} = e_{D_u} + e_{D_{uv}} + e_{D_v}.$$

For vuv , the three terms occur in the reverse order. Thus the translation parts also agree, proving the affine braid relations.

Each generator or inverse moves 0 a vector of norm one, so a word of length k moves 0 a distance at most k . For the linear permutation part r_s of s , the cocycle identity gives

$$V(t) - V(s) = r_s V(s^{-1}t).$$

Taking norms proves (4.3).

Reflection in D_u fixes its own basis vector, giving the translation vector $2e_{D_u}$ for u^2 . Also $(uv)^3 = (uvu)^2$. The affine map uvu has translation vector equal to the sum of the three mirror vectors, and its linear part preserves that sum and has square one. Squaring proves the second formula in (4.4). The third mirror at a vertex of an equilateral triangle is parallel to the opposite side and distinct from it. The three opposite directions are different. This proves the last assertion. $\square$

4.2 The parameter disk

For each pair $(u, v) \in \mathcal{E}$, use both commuting pairs

$$(U, H) = (u^2, z_{uv}), \quad (U, H) = (v^2, z_{uv}).$$

Commutation follows from Lemma 3.2. On the exponent plane $\mathbb{R}^2$ of each pair, let $\|\cdot\|_{U,H}$ be the stable seminorm from Lemma 2.3. Give its nonnegative quadrant exponent coordinates (λ, μ) , corresponding to U, H , respectively. The next lemma proves that this quadrant is a Euclidean sector.

Lemma 4.4. *Each of these six seminorms is positive definite. Each sector angle θ satisfies $0 < \theta < \pi/3$. The sectors glue isometrically along their shared rays, in the cyclic order*

$$a^2, z_{ab}, b^2, z_{bc}, c^2, z_{ca}, a^2. \quad (4.5)$$

Their radius-one disk P is a topological disk with total angle $0 < \Omega < 2\pi$. It has a continuous injective map $v_ : P \rightarrow E$ whose value in a sector for $U = u^2, H = z_{uv}$ is*

$$v_*(\lambda, \mu) = \lambda L_u + \mu L_{uv}, \quad (4.6)$$

and similarly with v in place of u .

Proof. For integral (i, j) , the pure translation formulas give

$$V((U^i H^j)^m) = m(iL_u + jL_{uv}).$$

Use (4.3) and (4.2), divide by m , and let $m \rightarrow \infty$. It follows that

$$\|iL_u + jL_{uv}\|_1 \leq C\|(i, j)\|_{U,H}.$$

The same inequality holds on rational vectors by homogeneity and on real vectors by continuity. The two displayed translation vectors are linearly independent: the extra-mirror coordinate first determines j , and the D_u -coordinate then determines i . The stable form is therefore positive definite.

The central pairing formula (3.4), with $r = 3$, gives

$$\langle U, H \rangle = \frac{\ell(H)^2}{3}.$$

Hence

$$\cos \theta = \frac{\langle U, H \rangle}{\ell(U)\ell(H)} = \frac{\ell(H)}{3\ell(U)} > \frac{1}{2}. \quad (4.7)$$

Positive definiteness ensures $\cos \theta < 1$. This proves the angle bounds.

Along a ray for an element W , exponent $t \geq 0$ has distance $t\ell(W)$ from the origin in either adjacent sector. Thus the identifications preserve both exponents and metrics. The resulting cone is parametrized by a radius $r \geq 0$ and a cyclic angle of period Ω , with all angle values identified at $r = 0$. Its part $0 \leq r \leq 1$ is the asserted disk.

The formulas for v_* agree on every shared ray. If a vector in its image has a nonzero extra-mirror coordinate, that coordinate specifies (u, v) and μ . The excess over 2μ on one of the original two side coordinates specifies the remaining exponent and which of the two sectors is used. If there is no excess, the point is on the shared H -ray. With no extra-mirror coefficient, the vector is on one squared-generator ray or is zero. These descriptions distinguish points precisely up to the stated ray identifications. They prove injectivity, including at the origin. $\square$

We use only the Euclidean metric within each sector of P ; we do not claim that this cone disk is CAT(0). Figure 1 shows the cyclic gluing.

4.3 Uniform asymptotic maps

In each sector define the group label

$$t_n(p) = U^{\lfloor n\lambda \rfloor} H^{\lfloor n\mu \rfloor}. \quad (4.8)$$

These labels agree on the glued rays.

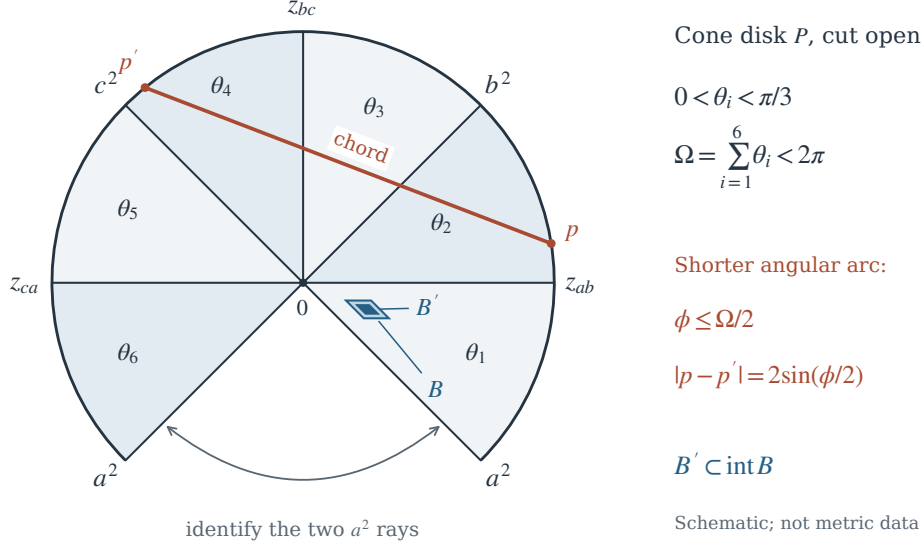


Figure 1: The six sectors of the parameter disk P , cut open along the a^2 -ray. The two boundary copies of that ray are identified. The angles and patch sizes are schematic; each angle is less than $\pi/3$, with total $\Omega < 2\pi$. The chord is drawn after unfolding a shorter angular arc and illustrates Lemma 4.6; the nested patches $B' \subset \text{int } B$ are chosen in Subsection 6.1 for the local winding count.

Lemma 4.5. *There are continuous maps $f_n : P \rightarrow Y$, with $f_n(0) = y$, and a constant C_0 independent of n, p , such that*

$$d_Y(f_n(p), t_n(p)y) \leq C_0. \quad (4.9)$$

For two points in any one sector,

$$\frac{d_Y(f_n(p), f_n(p'))}{n} \longrightarrow \|p - p'\|_{U,H} \quad (4.10)$$

uniformly. In addition, uniformly on P ,

$$\frac{d_Y(y, f_n(p))}{n} \longrightarrow r(p), \quad \frac{V(t_n(p))}{n} \longrightarrow v_*(p). \quad (4.11)$$

If $A_1, A_2 \subset P$ are disjoint nonempty closed sets, then for some $\delta > 0$ and all sufficiently large n ,

$$d_Y(f_n(A_1), f_n(A_2)) \geq \delta n. \quad (4.12)$$

Proof. Triangulate the entire exponent quadrant by squares of side $1/n$, each split along a diagonal, and assign the vertex $(i/n, j/n)$ the point $U^i H^j y$. Use constant-speed geodesics on edges. On each triangle, cone the opposite edge to its remaining vertex using geodesics. This gives a continuous map: all restrictions to common edges are the same geodesic parametrization. Geodesic convexity bounds the distance of every image point from a vertex by the maximum of the corresponding vertex distances. Commutation bounds those distances uniformly by the displacements of U, H . Restricting to the radius-one sector therefore gives (4.9). The restrictions on shared rays agree, so the six maps glue. There are only six sectors, allowing a common C_0 .

For one commuting pair put

$$F(i, j) = d_Y(y, U^i H^j y) \quad ((i, j) \in \mathbb{Z}^2).$$

The triangle inequality and commutation give

$$|F(i, j) - F(i', j')| \leq d(y, Uy)|i - i'| + d(y, Hy)|j - j'|.$$

For a fixed rational vector $q = (a/m, b/m)$, compare $\lfloor nq \rfloor$ with $\lfloor n/m \rfloor(a, b)$; their difference stays bounded. The definition of stable length consequently gives

$$\frac{F(\lfloor nq \rfloor)}{n} \longrightarrow \frac{\ell(U^a H^b)}{m} = \|q\|_{U, H}.$$

This includes negative a, b . On a bounded real region, choose a finite rational net of arbitrarily small mesh. The common Lipschitz bound above, with a rounding error $O(1/n)$, and continuity of the stable norm upgrade this convergence to uniform convergence on the region.

For p, p' in a sector, the difference between their floored exponent vectors differs from the floor of $n(p' - p)$ by a bounded integer vector. The orbit distance between their labels is F of that difference. The uniform result and (4.9) prove (4.10). Taking one point to be 0 gives the radial assertion. The other assertion of (4.11) follows directly from (4.4); its error is in fact $O(1/n)$.

Finally, compactness and injectivity give

$$\eta = \min_{p \in A_1, q \in A_2} \|v_*(p) - v_*(q)\|_1 > 0.$$

By (4.11), the norm of $V(t_n(p)) - V(t_n(q))$ is at least $\eta n/2$, uniformly for these pairs and large n . Combining (4.3), (4.2), and (4.9) gives

$$d_Y(f_n(p), f_n(q)) \geq \frac{\eta n}{2C} - 1 - 2C_0.$$

Decreasing the positive coefficient proves (4.12). $\square$

Lemma 4.6. *There is $\epsilon > 0$ such that, for all sufficiently large n , the map $f_n|_{\partial P}$ extends to a second parameter disk whose entire image lies outside $B_Y(y, \epsilon n)$. Gluing the two parameter disks gives a continuous sphere map, still denoted f_n .*

Proof. For any two points of ∂P , choose a shorter angular arc between them. Its angle ϕ is at most $\Omega/2 < \pi$. Unfold its sectors to a Euclidean wedge. The wedge intersected with the unit disk is convex, so the chord between the points stays within it, as in Figure 1. Its length is

$$2 \sin(\phi/2) \leq 2 \sin(\Omega/4).$$

Each sector appears at most once in this development, except that a sector containing both endpoints may appear in two end pieces. Each such piece is convex, so its intersection with the chord is a segment. The chord therefore splits into at most seven single-sector pieces. Apply the uniform estimate (4.10) to each piece. Uniformly for $p, p' \in \partial P$, this gives

$$d_Y(f_n(p), f_n(p')) \leq 2 \sin(\Omega/4)n + o(n). \quad (4.13)$$

Fix $p_0 \in \partial P$ and cone the boundary map to $f_n(p_0)$ by geodesics. Their continuous dependence makes this a continuous map of a parameter disk. If $q \in [A, B]$, the two triangle inequalities give

$$d(y, q) \geq \max\{d(y, A) - d(A, q), d(y, B) - d(q, B)\} \geq \frac{d(y, A) + d(y, B) - d(A, B)}{2}.$$

Here both endpoints have distance $n+o(n)$ from y , uniformly, by (4.11); (4.13) bounds their distance. The entire coning segment therefore stays at distance at least

$$(1 - \sin(\Omega/4))n - o(n)$$

from y . Take $\epsilon = \frac{1}{2}(1 - \sin(\Omega/4)) > 0$. □

5 Labelled triangular diagrams

We next construct a two-dimensional labelled complex from the presentation of T . The presentation and link are the case $m = n = p = 3$ of the construction of Brady and McCammond [2]. We give direct proofs of the needed facts. This complex is separate from the unknown CAT(0) space; the argument here concerns finite diagrams and planar topology.

We need two conclusions. Proposition 5.3 will show that a spherical diagram has zero total signed occurrences of each individual target triangle. Lemma 5.4 will approximate any geometric sphere map by a diagram, with one distance bound valid on every point of every parameter triangle. Together they will let the local winding count in Section 6 use the metric separation from the preceding section.

5.1 The triangular presentation and its link

Lemma 5.1. *For each $(u, v) \in \mathcal{E}$, introduce two private generators x_{uv} and h_{uv} . Write $(x_1, x_2, x_3, h) = (u, v, x_{uv}, h_{uv})$ within this block. A presentation of T is given by the three relations*

$$x_i x_{i+1} = h \quad (i = 1, 2, 3), \tag{5.1}$$

in each block, where the subscripts on the x_i are read modulo three.

Proof. The first two relations give $h = x_1 x_2$ and $x_3 = x_2^{-1} x_1 x_2$. Substituting these in the third gives $x_1 x_2 x_1 = x_2 x_1 x_2$. Conversely, that braid relation and these definitions of x_3, h give all three relations (5.1). Since the added generators are private to their blocks, eliminating them independently recovers the defining presentation of T . □

Let $\mathcal{A}$ be the resulting set of nine generators. Define the labelled Cayley complex $\mathcal{K}$ as follows. Its vertices are the elements of T . For every $g \in T$ and $q \in \mathcal{A}$ there is an oriented edge labelled (g, q) from g to gq . For each relation (5.1) and each g , attach a triangle, with its own cell identity, along the edges joining

$$g, \quad gx_i, \quad gh. \tag{5.2}$$

Edges and triangles retain these identities even if some other cells have the same vertex data. The homomorphism $T \rightarrow \mathbb{Z}$ taking a, b, c to 1 takes every x_i to 1 and every h to 2. Consequently the three vertices in (5.2) are distinct. In particular every target triangle has three distinct sides and three distinct corners.

The link at a vertex means the graph with one vertex for each incident edge direction and one edge for each incident triangle corner. Its vertices can be denoted q^+, q^- for $q \in \mathcal{A}$, representing the outgoing and incoming q -edges.

Lemma 5.2. *The link at every vertex of $\mathcal{K}$ has no circuit of length less than six, including circuits formed by parallel edges.*

Proof. Fix a Cayley vertex v . For the relation $x_i x_{i+1} = h$, its initial, middle, and terminal corners occur at v in exactly the translates $g = v$, $g = vx_i^{-1}$, and $g = vh^{-1}$, respectively. Their link edges are exactly

$$\{x_i^+, h^+\}, \quad \{x_i^-, x_{i+1}^+\}, \quad \{x_{i+1}^-, h^-\}. \quad (5.3)$$

Thus one block is a theta graph with three internally disjoint paths of length three from h^+ to h^- :

$$\begin{aligned} h^+ &- x_1^+ - x_3^- - h^-, \\ h^+ &- x_2^+ - x_1^- - h^-, \\ h^+ &- x_3^+ - x_2^- - h^-. \end{aligned}$$

Its nine link edges are distinct. The three blocks meet only at the signed directions of a, b, c . Across different blocks the only link edges whose two endpoints are shared directions are

$$\{a^-, b^+\}, \quad \{b^-, c^+\}, \quad \{c^-, a^+\}.$$

They too are distinct. This verifies both the complete link description and the absence of parallel edges; possible coincidences of group elements do not identify labelled edge directions.

It remains to bound a simple circuit. A circuit staying within one block has length at least six. Otherwise split it into maximal nonempty paths within individual blocks, and let k be their number. Their endpoints are the places where the circuit changes blocks. If $k = 2$, both changes occur in the intersection of the same two blocks. Their endpoints must be the two distinct signs of the one shared generator. Within either block their distance is three, so the circuit has length at least six.

If $k = 3$, the circuit visits all three blocks, changing at one signed direction of each of a, b, c . In a block the two relevant directions are those of x_1, x_2 . Equal signs have distance at least two; opposite signs have distance at least three except for x_1^-, x_2^+ , which are adjacent. When all three signs agree, the length is therefore at least $2 + 2 + 2$. Otherwise there are two changes of sign in the cyclic order a, b, c, a : one from minus to plus, and one from plus to minus. At most one of the corresponding two block paths can use the exceptional adjacency. The lower bound is then $1 + 3 + 2 = 6$.

Finally, if $k \geq 4$, a block path of length one must be one of the three displayed edges between shared directions. Each shared direction is incident to at most one such edge, so two length-one paths cannot be consecutive on the simple circuit. At most $\lfloor k/2 \rfloor$ paths have length one and the others have length at least two. The total length is at least $2k - \lfloor k/2 \rfloor \geq 6$. $\square$

5.2 Signed counts on spheres

A combinatorial map to $\mathcal{K}$ will always include cell identities. Each edge maps to a specified target edge, possibly with its orientation reversed, or to a vertex. Each triangle either maps homeomorphically to a specified target triangle by its vertex correspondence, or maps affinely into a specified target edge or vertex. These choices agree on common sides. We call the latter triangles collapsed. Fix an orientation for every target triangle. On an oriented source surface, a noncollapsed triangle has sign $+1$ or -1 according to its map to that oriented target.

The proof below uses the classical diagrammatic-reducibility method of deleting opposite faces and excluding a reduced spherical diagram by curvature [11, Definition 3.1 and Corollary 4.6], here keeping the identity of each target cell and allowing collapsed parameter triangles.

Proposition 5.3. *For every combinatorial map from a finite triangulated oriented two-sphere to $\mathcal{K}$, and every individual target triangle τ , the sum of the signs of all occurrences of τ is zero. Collapsed triangles are allowed in the map and contribute zero.*

Proof. Encoding the sphere. In each noncollapsed source triangle draw a central node with three legs ending at its side midpoints. A triangle collapsed to an edge has exactly two nonconstant sides. Join their midpoints by an arc; both sides map to the same labelled target edge. A triangle collapsed to a vertex needs no arc. Joining across source sides produces an embedded trivalent graph, together with possible closed curves containing no nodes. Discard these curves. Each remaining node records a particular occurrence of a target triangle, with its orientation sign, and each graph edge pairs two sides mapping to the same labelled target edge.

We must retain endpoint information before using this graph as a diagram. Give it a thin regular neighbourhood, consisting of node disks and ribbons along the edges. Along either side of a ribbon the actual group label of the corresponding endpoint of its target edge stays constant. Indeed, passing through an edge-collapsed triangle connects the two sides at the same endpoint of that edge. At a node the two ribbon sides bounding one sector carry the same group label, namely the corresponding corner of that target triangle. It follows that every boundary circle of the regular neighbourhood carries one group-vertex label throughout.

For each connected component containing nodes, cap its neighbourhood by a disk on every boundary circle and use that constant label at the corresponding dual vertex. This gives a labelled triangular cell decomposition of a sphere: graph nodes correspond to its triangular faces, graph edges to its edges, and neighbourhood boundary circles to its vertices. Here we use the elementary finite planar-topology fact that a polygonal arc properly embedded in a disk cuts it into two disks. To see how it applies, thicken a spanning tree of the connected planar graph to a disk. Add the other embedded edges one at a time as bands. Each runs through a complementary disk and divides it into two disks. Capping the remaining boundary circles therefore gives a sphere. This also proves the usual Euler formula for the resulting cell decomposition. It need not be a strict simplicial complex; incidences, rather than distinct pairs of vertices, are counted.

The labelling rules exclude certain possible degeneracies. A graph edge cannot join two legs at the same node: these legs represent different target sides, whereas a ribbon has one fixed target edge. No two corners of a face can become the same dual vertex, since their actual target vertex labels are distinct. Also a bridge of the graph would have both ribbon sides on the same neighbourhood boundary circle. Constancy around that circle would equate the distinct endpoints of its target edge, so a correctly labelled graph here has no bridge. Multiple edges between different nodes are harmless.

Deleting folds. A fold is a pair of faces sharing an edge and mapping to the same target triangle across the same target side. Their signs are opposite. Such a pair can be deleted directly in the planar graph. Take a small disk around the chosen connecting edge and its two nodes, cutting the other four legs close to the nodes. The chosen edge and its endpoint disks form a disk even if the same two nodes have additional connecting edges: the middle portions of those additional edges lie outside this disk. If the remaining target sides are called i, j , the exposed legs occur in cyclic order i, j, j, i , up to reversal and cyclic permutation. Join i to i and j to j inside the disk without crossing and delete the two nodes. The endpoint labels match on each new ribbon side, including at the common target corner between the two pairings. This preserves all the label rules, planarity, and the sum of signs for each target triangle. It may separate components or produce circles without nodes; discard such circles and cap the neighbourhoods of the remaining components as above. Figure 2 illustrates the reconnection and its endpoint labels.

The Euler contradiction. If nodes remain and no fold is available, choose a component and its dual spherical cell decomposition. The successive corners at each dual vertex give a closed walk in the link of its actual target vertex. A backtrack in this walk would use the very same target corner on both sides of a shared edge. The exact corner description (5.3) then identifies the same translated

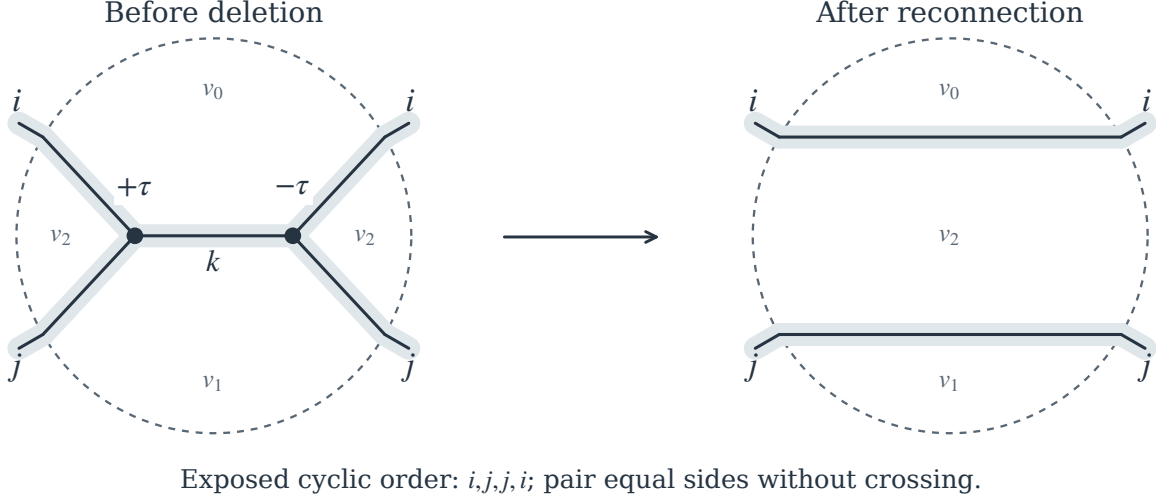


Figure 2: Deleting a fold. The two nodes are oppositely signed occurrences of the same labelled target triangle τ , joined across side k . Its other sides are i, j ; the regions retain the actual vertex labels v_0, v_1, v_2 . Inside the dashed surgery disk, equal sides can be reconnected without crossing, preserving all endpoint labels.

target triangle on both sides, which is a fold. These must be two distinct source faces: one face cannot use two different corners at this dual vertex, nor can two of its distinct sides be joined by the same ribbon. Thus a one-face incidence cannot evade the deletion rule.

Every link walk is consequently nonbacktracking. Any finite nonempty closed nonbacktracking walk contains a circuit, so Lemma 5.2 makes its length at least six. Let F, E, V count faces, edges, and vertices of this dual component, with all incidences counted. We obtain

$$3F = 2E, \quad 2E \geq 6V, \quad V - E + F = 2.$$

The first two imply $V - E + F \leq 0$, a contradiction. Thus whenever nodes remain, a fold can be removed. Each removal decreases their number by two, so the process terminates with no nodes. All signed target-cell counts have been preserved and are then zero. $\square$

5.3 Controlled approximation by diagrams

The next lemma transfers continuous maps into the geometric T -space to diagrams. A triangulation of a parameter surface means a homeomorphism from a finite triangulated surface onto it. Refinements below are performed within its existing faces, with matching subdivisions of their boundaries. In particular, a prescribed subcomplex remains a union of closed triangles after refinement; its underlying subset is never moved.

Lemma 5.4. *Suppose that T acts geometrically on Y and fix $y \in Y$. There is a constant C_1 , depending only on this action, y , and the fixed presentation, with the following property. Let Σ be a triangulated parameter two-sphere, let B be a subcomplex, and let $f : \Sigma \rightarrow Y$ be continuous. There is a refinement keeping B a subcomplex and a combinatorial map from the refined sphere to $\mathcal{K}$ such that, for every final triangle Q , every one of its group-vertex labels s , and every $p \in Q$,*

$$d_Y(sy, f(p)) \leq C_1. \tag{5.4}$$

In particular C_1 is independent of f , its modulus of continuity, and the number of triangles required.

Proof. Uniform boundary types. Choose $R < \infty$ such that every point of Y is within R of Ty . For example, if $Y = TK$ with K compact, take $R = \max_{z \in K} d_Y(y, z)$. By Lemma 4.2, choose an integer M such that

$$d_Y(sy, ty) \leq 2R + 1 \implies d_T(s, t) \leq M. \quad (5.5)$$

Set $D = \max\{d_Y(y, ay), d_Y(y, by), d_Y(y, cy)\}$.

By uniform continuity, first refine the given parameter triangulation, still respecting B , until every face image has diameter at most one. Give each vertex v a group label s_v with $d_Y(s_v y, f(v)) \leq R$. For each unoriented edge, choose once and for all a path between its two labels of length at most M in the Cayley graph on a, b, c ; the opposite orientation uses the reverse of this same path. This is possible by (5.5). Add three constant steps at each end of every chosen path. Subdivide the parameter edge accordingly, once for both adjacent faces. Its length is at most $M + 6$, and every face boundary is a labelled null loop of length at most $3(M + 6)$.

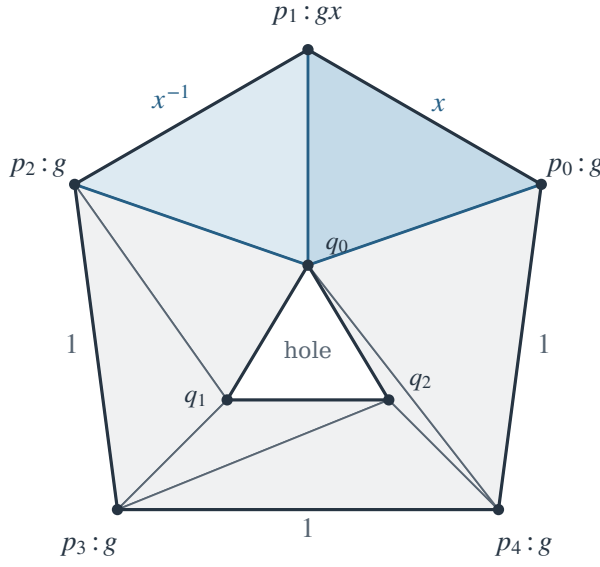
Relative to the label of an original vertex of a face, only finitely many such boundary types are possible: their lengths are bounded and their letters belong to $\{a^{\pm 1}, b^{\pm 1}, c^{\pm 1}, 1\}$. We now construct one finite filling of each type, keeping its parameter boundary fixed. This construction also explains why an ordinary parameter disk, with permissible collapsed triangles, can always be used.

Filling the boundary loops. Reserve the first three constant steps at the chosen original base vertex, and leave them unchanged. Ignoring constants, the remaining boundary word represents the identity in the presentation of Lemma 5.1. By the definition of a presented group, it lies in the normal closure of the relators in the free group. Hence it can be reduced to the empty word by finitely many free insertions or deletions of qq^{-1} , insertions or deletions of a translated triangular relator loop or its reverse, and constant-step moves. Here each individual move takes place in one target edge, triangle, or vertex. More explicitly, an insertion of a conjugate wrw^{-1} is achieved by inserting the journey ww^{-1} through successive free moves and then inserting the triangular loop r at its tip. Free cancellations account for equality in the free group. This proves the claimed finite reduction without assuming a disk-diagram theorem. All moves are performed outside the three reserved steps, so every intermediate cyclic parameter path has at least three edges.

Realize two successive cycles as the disjoint boundary circles of a triangulated annulus. Along unchanged edges use strips whose two ends have identical target labels on the two circles; triangulate each strip rectangle into two edge-collapsed or vertex-collapsed triangles. For a nonempty substituted arc on each circle, the intervening region is a polygon whose vertex labels lie in the one target simplex used by the move, so it can be triangulated with the corresponding simplex maps.

For precision, an empty arc requires a fan, not an identification of parameter vertices. Suppose an arc with successive vertices $p_0, \dots, p_k$ on one circle is deleted at the vertex q of the other. The region between these arcs has the two distinct radial edges qp_0 and qp_k , together with the fan triangles $qp_{j-1}p_j$ for $1 \leq j \leq k$. Although p_0, p_k, q have the same group label, p_0 and p_k are distinct parameter vertices. The unchanged strips attach along the two radial edges. At q the preceding strip, the fan, and the following strip have a single interval as their link, so this construction has no pinched parameter point. Reversing the construction handles insertion at an empty arc. The reserved constant steps ensure a nonempty unchanged part around the other side of the annulus. Thus all these pieces form an actual annulus with its two boundary circles unchanged. Since all vertices of the fan map to the simplex used by the move, its triangles are legal, including when some or all of their labels agree.

As a concrete case, deleting xx^{-1} from a five-edge cycle consisting of that excursion and three constant edges gives a three-edge cycle. The connecting annulus has two fan triangles collapsed to the x -edge and three unchanged strips. Its inner circle remains a circle, despite all its labels being equal; see Figure 3.



Delete xx^{-1}

Five edges $\rightarrow$ three edges

q_0, q_1, q_2 all have label g .

Blue fan: two triangles
mapping into the edge $g-gx$.

Gray strips: six triangles
mapping to the vertex g .

The inner cycle is capped later.

Figure 3: The annulus deleting xx^{-1} and retaining three constant steps. The five outer labels are g, gx, g, g, g ; the three inner labels are all g . The two fan triangles map into the edge from g to gx ; the six triangles in the other three strips map to g . The inner boundary is capped only at the end of the reduction.

Stack the finitely many annuli and cap the final three-step constant cycle by a cone mapping to its vertex. The result is a finite triangulated disk with the original subdivided boundary. Identify this parameter disk with the original face by a boundary-preserving homeomorphism. Such an identification is elementary: realize the stacked annuli as successive collars inside a disk and the final cone as its central disk; a prescribed parametrization of the outer circle extends radially. The triangulation and its affine simplex maps are understood in these abstract triangle coordinates; the homeomorphism need not be affine in any pre-existing coordinates on the parameter surface. All added triangles are therefore contained in the original face. Gluing these constructions along the already agreed edge subdivisions gives a triangulated sphere and preserves B as a subcomplex.

The uniform estimate. It remains to verify the uniform estimate at every parameter point. Fix all the above fillings for the finite set of boundary types. Normalize their original base vertex to $1 \in T$, and let L be the largest original word length $d_T(1, s)$ of any vertex label appearing in any one of these fillings. This number is finite, including for private generators in the triangular presentation, since they are words in a, b, c . It is chosen once and does not depend on f . In a translated filling based at an original vertex v_0 labelled s_0 , every label s satisfies $d_T(s_0, s) \leq L$. Every point p of that original face then satisfies

$$\begin{aligned} d_Y(sy, f(p)) &\leq d_Y(sy, s_0y) + d_Y(s_0y, f(v_0)) + d_Y(f(v_0), f(p)) \\ &\leq DL + R + 1. \end{aligned}$$

Each final triangle lies within that face, so taking $C_1 = DL + R + 1$ proves (5.4). $\square$

6 A local winding count

We now combine the geometry of Section 4 with the diagram statements of Section 5. Continue to assume the action and strict inequalities of Proposition 4.1, and retain the sphere maps f_n and the constant $\epsilon > 0$ from Lemma 4.6.

Write Σ for the parameter sphere; for the geometric map $f_n : \Sigma \rightarrow Y$, Lemma 5.4 supplies a separate labelled diagram $\Sigma \rightarrow \mathcal{K}$ on that sphere. Projected mirror coordinates of its group-labelled vertices, extended affinely over parameter triangles, will define a planar map $q_n : \Sigma \rightarrow \mathbb{R}^2$ that approximates coordinates on a small patch and has nonzero local winding. Separation in Y will force every occurrence of a contributing target triangle into the patch, making that winding a weighted sum of whole-sphere signed counts, which Proposition 5.3 makes zero.

6.1 A patch and its coordinates

Choose one of the six sectors, say the sector for $U = u^2, H = z_{uv}$. Its exponent coordinates form the nonnegative quadrant in a Euclidean plane. Choose a small closed coordinate rectangle

$$B = [\lambda_0, \lambda_1] \times [\mu_0, \mu_1]$$

in the interior of this sector, with $\lambda_0, \mu_0 > 0$, and so small and close to the origin that all its radii satisfy

$$0 < r(p) < \min(\epsilon/3, 1/4) \quad (p \in B). \quad (6.1)$$

Write $\kappa(p) = (\lambda, \mu)$ on B . Fix an interior point p_* and a smaller closed coordinate rectangle $B' \subset \text{int } B$ containing p_* in its interior.

We regard P and the second disk as a fixed triangulated parameter sphere, with B a subcomplex. One explicit way to arrange this is as follows. In each sector take the triangle whose two outer vertices lie on its boundary rays at radius one. The six triangles form a polygonal cone disk Q . On a radial line, its outer radius ρ is at least $\cos(\theta/2) > \sqrt{3}/2$. Map Q to P by a radial homeomorphism that fixes radii at most $1/2$ and is linear from radius $1/2$ to the outer radius, sending ρ to 1. These maps agree on the common rays. The rectangle B is fixed pointwise. Extend its four straight sides within its triangle to partition that triangle into finitely many convex polygons; triangulate each polygon and the other five triangles. Finally cone the resulting boundary subdivision to a new parameter vertex for the second disk. This supplies the required finite parameter triangulation.

For $\xi = \sum_D \xi_D e_D$, define the bounded linear map

$$J\xi = \left(\frac{\xi_{D_u} - \xi_{D_{uv}}}{2}, \frac{\xi_{D_{uv}}}{2} \right). \quad (6.2)$$

By (4.4),

$$Jv_*(p) = \kappa(p) \quad (6.3)$$

throughout the chosen sector. This is a coordinate projection on that sector; no global injectivity of J is asserted.

Apply Lemma 5.4 to each sphere map f_n , preserving B as a subcomplex. We obtain a subdivided combinatorial sphere map to $\mathcal{K}$. Denote the group label at a vertex by s . There is one constant C_1 , independent of n , for which

$$d_Y(sy, f_n(p)) \leq C_1 \quad (6.4)$$

whenever s is a vertex label of a final parameter triangle and p is any point of that triangle.

On this parameter sphere, assign to the vertex with label s the plane value $JV(s)/n$, and extend affinely on each abstract triangle. Call the resulting continuous map q_n . Affineness refers to the abstract triangle coordinates, regardless of the homeomorphisms used to place the triangles in the parameter sphere.

Lemma 6.1. *On B , the maps q_n converge uniformly to κ .*

Proof. Fix $p \in B$ and any vertex label s of a triangle containing p . The bounds (6.4) and (4.9) imply

$$d_Y(sy, t_n(p)y) \leq C_1 + C_0.$$

By (4.2) and (4.3), there is a constant C_2 , independent of n, p, s , such that

$$\|V(s) - V(t_n(p))\|_1 \leq C_2.$$

The second convergence in (4.11) has error $O(1/n)$, so

$$\left\| \frac{V(s)}{n} - v_*(p) \right\|_1 = O(1/n)$$

uniformly for every such vertex label. Apply J and (6.3). Since $q_n(p)$ is a convex combination of those vertex values, it satisfies the same uniform error bound from $\kappa(p)$. This is precisely where the all-points form of (6.4) is used. $\square$

6.2 The winding calculation

For completeness, we spell out the elementary planar count we use. For a loop avoiding a point w , choose continuous local arguments of its vectors from w on successive short parameter intervals. Matching the arguments at their common endpoints gives a total angle change in $2\pi\mathbb{Z}$. Its quotient by 2π is the winding number. It is unchanged under a homotopy avoiding w : local angle choices vary continuously, and a continuous integer-valued function is constant.

If a finite oriented triangulated disk is mapped affinely to the plane, and w avoids all projected edges, sum the winding numbers of its oriented triangle boundaries. Interior edges occur twice with opposite orientations and their angle increments cancel. The sum is therefore the winding number of the outer boundary. For one affine triangle the term is zero if w is outside its convex hull, and is its orientation sign if w lies in its interior. In the first case one may contract the loop inside its convex hull; in the second, rays from w meet the triangle boundary once and its direction makes one full turn. A degenerate triangle contributes zero because w avoids its projected edges.

Choose a small open plane ball W about $\kappa(p_*)$ whose closure lies in $\kappa(\text{int } B')$. By Lemma 6.1, for large n and every w in a sufficiently small fixed such ball, the following hold:

1. the straight-line homotopy from $q_n|_{\partial B}$ to $\kappa|_{\partial B}$ avoids w ;
2. every $p \in B$ with $q_n(p) = w$ lies in $\text{int } B'$.

Indeed, take the approximation error and the radius of W smaller than half the positive distance from $\kappa(p_*)$ to $\kappa(B \setminus \text{int } B')$. For each large n , choose $w \in W$ avoiding every projected edge of the entire finite sphere diagram. A finite union of line segments cannot contain an open ball, so this choice is possible.

The positively or negatively oriented rectangle $\kappa(\partial B)$ has winding number 1 or -1 about w . The preceding homotopy and cancellation argument show that the sum of signed coverings of w by the triangles of B is nonzero. Every contributing triangle is noncollapsed, has a nondegenerate planar image, and contains a parameter point of B' projecting to w .

6.3 All occurrences of a contributing cell are local

Lemma 6.2. *For all sufficiently large n , every occurrence in the sphere diagram of a target triangle contributing to the winding count on B lies entirely in $\text{int } B$.*

Proof. By (4.12), the f_n -images of B' and $P \setminus \text{int } B$ are separated by a positive multiple of n . On the second disk, Lemma 4.6 gives distance at least ϵn from y . On B' , (6.1) and (4.11) give distance at most $(\epsilon/3)n + o(n)$ from y . Thus the images of B' and the entire complement of $\text{int } B$ in the sphere have mutual distance tending to infinity, uniformly.

Let τ be a contributing labelled target triangle. One of its occurrences contains $p \in B'$ with $q_n(p) = w$. Fix any one of the three group vertex labels s of τ . Every other occurrence of this same target cell has the same vertex label s , including the full translate information. If some occurrence met the complement of $\text{int } B$, take a point p' in that intersection. The all-points bound (6.4) would give

$$d_Y(f_n(p), f_n(p')) \leq d_Y(f_n(p), sy) + d_Y(sy, f_n(p')) \leq 2C_1,$$

contradicting the uniform separation for large n . This also excludes an occurrence touching ∂B . $\square$

Proof of Proposition 4.1. Choose n large enough for the constructions and Lemma 6.2. Orient every labelled target triangle of $\mathcal{K}$ once and for all. Its three group vertices have fixed projected values $JV(s)/n$, so its affine planar image is fixed. If it covers w , its determinant has a fixed nonzero sign $\eta_\tau \in \{1, -1\}$. The covering sign of each occurrence is η_τ times that occurrence's orientation sign relative to the chosen orientation of the target.

Group the nonzero winding sum on B by its contributing target triangles. By Lemma 6.2, every occurrence of each of these target triangles in the whole sphere lies in B . Its total contribution is therefore

$$\eta_\tau (\text{signed number of occurrences of } \tau \text{ in the sphere}) = 0$$

by Proposition 5.3. Summing gives zero, contrary to the winding number 1 or -1 . Distinct target cells remain separate in this sum even if their projected vertex data agree. The contradiction proves the proposition. $\square$

7 The finite Artin matrix and the final contradiction

We now combine the length restriction of Proposition 3.1 with the obstruction of Proposition 4.1. The additional generators in the following matrix make the group T a common centralizer. This will produce a geometric action of T from any hypothetical geometric action of the full group.

Definition 7.1. Let $\mathcal{E} = \{(a, b), (b, c), (c, a)\}$, as before, and set

$$S = \{a, b, c, d, e\} \sqcup \{s_j^{uv} : (u, v) \in \mathcal{E}, 3 \leq j \leq 10\}.$$

All symbols in the second set are distinct. For each $(u, v) \in \mathcal{E}$, also write $s_1^{uv} = u$ and $s_2^{uv} = v$; these are aliases for existing elements of S . Define a symmetric matrix $M = (m_{st})_{s, t \in S}$ by the following prescriptions:

1. $m_{ss} = 1$ for every $s \in S$, and $m_{ab} = m_{bc} = m_{ca} = 3$.
2. In each block $\{s_1^{uv}, \dots, s_{10}^{uv}\}$, set

$$m_{s_i^{uv}, s_j^{uv}} = \begin{cases} 3, & |i - j| = 1, \\ 2, & |i - j| \geq 2. \end{cases}$$

3. For $q \in \{d, e\}$ and $u \in \{a, b, c\}$, set $m_{qu} = 2$.

4. Every remaining off-diagonal entry is ∞ .

Let $G = A_M$ be the resulting Artin group, with the generators denoted by the elements of S themselves. Let G_0 be the group presented by the generators $S \setminus \{d, e\}$ and all their prescribed relations.

Two distinct blocks meet in just one generator, so their prescriptions do not conflict. Each pair among a, b, c is the initial pair of exactly one block, with label 3 in both prescriptions. Thus the definition specifies exactly one matrix, on

$$|S| = 5 + 3 \cdot 8 = 29$$

generators. In particular $m_{de} = \infty$, and neither d nor e has a finite label with any of the private block generators.

7.1 The required embeddings

Lemma 7.2. *For every $(u, v) \in \mathcal{E}$, the homomorphism*

$$B_3 = \langle t_1, t_2 \mid t_1 t_2 t_1 = t_2 t_1 t_2 \rangle \longrightarrow T, \quad t_1 \longmapsto u, \quad t_2 \longmapsto v,$$

is injective. For every $l \geq 3$, the homomorphism $B_3 \rightarrow B_l$ given by $t_1 \mapsto s_1$ and $t_2 \mapsto s_2$ is also injective.

Proof. For the first assertion, let w be the remaining element of $\{a, b, c\}$. The assignment

$$u \longmapsto t_1, \quad v \longmapsto t_2, \quad w \longmapsto t_1$$

defines a homomorphism $T \rightarrow B_3$: the braid relation for u, v is the displayed defining relation of B_3 , the relation for v, w is that relation with its sides interchanged, and the relation for w, u becomes $t_1^3 = t_1^3$. Its composite with the asserted inclusion is the identity of B_3 .

For the second assertion we construct a word invariant, retaining the entire permutation of the strands as part of its state. Place l distinct tokens, named $1, \dots, l$, in l ordered positions. Tokens $1, 2, 3$ are called selected. A state consists of an ordered list of all l tokens and an element $q \in B_3$. Read a signed letter s_i^ε , $\varepsilon \in \{1, -1\}$, as follows. Swap the tokens in positions $i, i+1$, irrespective of the sign. If both are selected, they are consecutive in the ordered list of selected tokens; if they occupy positions $j, j+1$ in that shorter list, replace q by qt_j^ε . If at most one is selected, leave q unchanged. The integer j is computed before the swap, and is either 1 or 2.

We verify the defining relations as transformations of all states, not merely of the initial state. The operation for s_i^{-1} is the inverse of that for s_i : a second swap restores the full token list, and, when a letter was recorded, its selected-list index is still j , so that the two recorded letters cancel. For $|i - k| \geq 2$, the swaps at i and k involve disjoint pairs of positions and have the same final token list in either order. With just three selected tokens, at most one of these pairs consists of two selected tokens. Swapping the other pair does not change the selected-list index of the recorded pair, because moving a selected token past an unselected token preserves the order of the selected list. Hence the positive commutation relation is respected.

For a positive braid relation, name the three tokens initially in positions $i, i+1, i+2$ by A, B, C . The swaps in $s_i s_{i+1} s_i$ exchange, in order, the pairs

$$(A, B), (A, C), (B, C),$$

whereas those in $s_{i+1}s_is_{i+1}$ exchange them in the opposite order. Both finish with the ordered triple C, B, A and fix all other positions. If at most one of A, B, C is selected, neither word records a letter. If exactly two are selected, each word records their single crossing with positive sign and the same selected-list index: the third selected token is outside these three consecutive positions and stays on the same side throughout. If all three are selected, the two records are respectively $t_1t_2t_1$ and $t_2t_1t_2$, which represent the same element of B_3 .

Thus the invertible state transformations satisfy every defining positive relation of B_l . They consequently respect arbitrary signed word equalities: each defining relator acts trivially, as does its inverse or any conjugate, and free cancellations act trivially by the inverse check. This is precisely the normal-closure description of equality in the presented group. In particular, the verification applies after every signed prefix, and the equal full token lists ensure identical processing of every suffix.

Start with the token list $(1, \dots, l)$ and $q = 1$. A word using only $s_1^{\pm 1}, s_2^{\pm 1}$ always swaps two of the first three tokens. Its record is exactly the same word with s_i replaced by t_i , including every sign. If the original word is trivial in B_l , its state transformation fixes this initial state, and its record is therefore trivial in B_3 . This proves injectivity. The construction asserts a state invariant; it does not assert that discarding the other tokens gives a homomorphism $B_l \rightarrow B_3$. $\square$

7.2 Normal forms and the common centralizer

We use the standard normal-form theorem for free products with amalgamation; see Serre [17, Chapter I, §1.2, Theorems 1–2]. The elementary proof below records the exact form needed here, including the case in which multiplication removes the final representative.

Lemma 7.3. *Let H_1, H_2 be groups with a common embedded subgroup E , and let $H_1 *_E H_2$ be the group obtained by identifying the two copies of E in their free product. Each factor embeds in this group, and their intersection is E . A product $a_1 \cdots a_n$, $n \geq 1$, whose factors belong alternately to $H_1 \setminus E$ and $H_2 \setminus E$ is nontrivial. If such a product contains a factor from $H_i \setminus E$, it does not belong to the other factor.*

Proof. For $i = 1, 2$, choose a set $R_i \subset H_i$ such that every element of H_i is uniquely re with $r \in R_i$, $e \in E$, and with 1 representing E . Consider formal states

$$(r_1, \dots, r_n; e), \quad n \geq 0,$$

where every r_j is a nonidentity representative and their factor types alternate; for $n = 0$ only the terminal element e is present. Representatives retain their factor types as part of the state.

Define right multiplication of a state by $h \in H_i$ as follows. If its final representative has type i , isolate the tail re with that representative r ; otherwise use the missing representative $r = 1$. The preceding list P is empty or ends in the other factor. Decompose the element reh of H_i uniquely as $r'e'$. Replace the tail by $r'; e'$, omitting r' when $r' = 1$.

For fixed i , these operations are a right action of H_i . Indeed, after multiplying by $h \in H_i$, the prefix P stays fixed. For a further $k \in H_i$, the tail is rewritten from $r'e'k = (reh)k$. If $r' = 1$ was omitted, the rule simply uses the missing representative 1 and rewrites $e'k$, with exactly the same result. Uniqueness of the decomposition in H_i and associativity therefore give the operation for hk . Multiplication by 1 fixes every state, and multiplication by h^{-1} is inverse to multiplication by h .

If $h \in E$, either factor action leaves all representatives fixed and changes only the terminal element from e to eh . The two actions therefore agree on E and give a right action of the presented amalgam: all multiplication relations within the factors and all identifications of E act identically.

Every state is reached from $(;1)$ by reading the word $r_1 \cdots r_n e$. Conversely, applying the algorithm to an arbitrary word uses only valid equalities in the amalgam and produces a word of that form. If two state words were equal in the amalgam, their actions on $(;1)$ would give the same state. This proves both existence and uniqueness of these normal forms. In particular, the normal form of an element of H_i is its single representative and terminal E -element, or just its E -element; this proves the factor embeddings and the intersection assertion.

Finally, when processing an alternating product $a_1 \cdots a_n$ with every $a_j \notin E$, at each step the terminal E -element times a_j is still outside E . Since its factor type differs from that of the preceding representative, one new nonidentity representative is appended. Its normal form thus has exactly n representatives, of the indicated alternating types. It is nontrivial and cannot be the normal form of an element of a factor whose type omits any of these representatives. $\square$

Write B_{11}^{uv} for the abstract braid group in the block corresponding to (u, v) , and B_3^{uv} for its initial two-generator subgroup. By Lemma 7.2, each B_3^{uv} embeds both in that block and in T . Applying Lemma 7.3 successively therefore gives

$$G_0 \cong ((T *_{B_3^{ab}} B_{11}^{ab}) *_{B_3^{bc}} B_{11}^{bc}) *_{B_3^{ca}} B_{11}^{ca}. \quad (7.1)$$

At every stage T remains embedded, so the next attaching subgroup is embedded as well. The presentations agree: identifying the first two generators of each block with its specified pair in T leaves exactly the generators and relations defining G_0 . Thus T and all three braid blocks are embedded in G_0 .

Let $F(d, e)$ be the free group on d, e , realized as reduced words in $d^{\pm 1}, e^{\pm 1}$ with multiplication by concatenation and free cancellation, and put $B = T \times F(d, e)$. The direct product has the presentation for T together with the two free generators and the six commutation relations between $\{d, e\}$ and $\{a, b, c\}$. To check this description directly, the commutations move every T -letter to the left of every free letter; the evident homomorphism to the direct product and the two commuting factor homomorphisms are inverse on the generators. Comparison with Definition 7.1 now gives

$$G \cong G_0 *_T B = G_0 *_T (T \times F(d, e)). \quad (7.2)$$

Lemma 7.3 makes both factors, and hence T and the three braid blocks, embedded subgroups of G .

Lemma 7.4. *For the elements d, e of G ,*

$$C_G(d) \cap C_G(e) = T.$$

Proof. Every element of T commutes with d and e by the presentation. Conversely, first suppose that $g \notin B$. Its normal form in (7.2) can be written $g = wb$, where $b \in B$ and w is a nonempty alternating product of elements outside T , whose last factor lies in $G_0 \setminus T$. Indeed, absorb the terminal T -element and, if present, the last B -representative into b . There is at least one G_0 -representative because $g \notin B$.

Under the projection $B \rightarrow F(d, e)$, the element bdb^{-1} maps to a conjugate of the nonidentity free generator d . Consequently $bdb^{-1} \in B \setminus T$. The word

$$gdg^{-1} = w(bdb^{-1})w^{-1}$$

is reduced in the sense of Lemma 7.3: its middle factor lies in $B \setminus T$, both neighbors lie in $G_0 \setminus T$, and all remaining adjacent factors alternate. It contains a $G_0 \setminus T$ factor and hence cannot belong to B . In particular it cannot equal d . Thus $C_G(d) \subset B$.

It remains to compute the common centralizer within B . If a reduced word $f \in F(d, e)$ commutes with d , write $f = vd^q$, where $q \in \mathbb{Z}$ and either v is empty or its last letter is e or e^{-1} . This is obtained by removing the maximal terminal string of d -letters from the reduced word. Then

$$fdf^{-1} = vdv^{-1}.$$

If v is nonempty, this displayed word is freely reduced, has length $2|v| + 1 > 1$, and cannot equal d . Hence $f = d^q$. For $q \neq 0$, the word $d^q ed^{-q}$ is freely reduced and is not e , so only the identity commutes with both free generators. An element $(t, f) \in T \times F(d, e)$ therefore centralizes both d and e exactly when $f = 1$. This proves the assertion. $\square$

7.3 A geometric action of the common centralizer

Geometric actions of centralizers on minimum-displacement sets are classical. Swenson [18, Theorem 13] credits this result to Ruane [16]; Swenson also studies intersections of convex subgroups, and his proof of Theorem 17 treats simultaneous centralizers [18, Theorems 16–17]. The following finite-tuple compactness argument gives the precise common-displacement-sublevel formulation needed here. It applies in every dimension and does not require the elements being centralized to commute.

Lemma 7.5. *Suppose a group H acts geometrically on a nonempty proper $CAT(0)$ space X . For finitely many elements $q_1, \dots, q_m \in H$, $m \geq 1$, set*

$$C = \bigcap_{j=1}^m C_H(q_j).$$

If $A \geq 0$ is finite and the set

$$Y_A = \{x \in X : d_X(x, q_j x) \leq A \text{ for } 1 \leq j \leq m\}$$

is nonempty, then Y_A , with its inherited metric, is a proper $CAT(0)$ space, invariant under C , and the restricted C -action is geometric. Every element of C has the same stable translation length in Y_A as in X .

Proof. Each function $x \mapsto d_X(x, q_j x)$ is continuous and convex. For the convexity, if γ is the geodesic from x_0 to x_1 parametrized on $[0, 1]$, then $q_j \gamma$ is the geodesic from $q_j x_0$ to $q_j x_1$, and Lemma 2.1 gives

$$d_X(\gamma(t), q_j \gamma(t)) \leq (1-t)d_X(x_0, q_j x_0) + td_X(x_1, q_j x_1).$$

Thus Y_A is closed and convex. Its geodesic triangles and their metrics are inherited from X , so it is $CAT(0)$; each closed bounded subset of Y_A is closed and bounded in X and is therefore compact. If $c \in C$, then

$$d_X(cx, q_j cx) = d_X(cx, cq_j x) = d_X(x, q_j x),$$

so Y_A is C -invariant. Compact subsets of Y_A are compact in X , hence the restricted action satisfies the required compact-set properness condition.

It remains to prove cocompactness. Choose compact $K \subset X$ with $HK = X$, and let

$$L = \{z \in X : d_X(z, K) \leq A\}.$$

This closed bounded neighborhood of K is compact, since X is proper. Accordingly

$$\mathcal{F} = \{r \in H : rL \cap L \neq \emptyset\}$$

is finite. For any $h \in H$ such that $hK \cap Y_A \neq \emptyset$, choose $p \in K$ with $hp \in Y_A$. For every j ,

$$d_X(p, h^{-1}q_jhp) = d_X(hp, q_jhp) \leq A.$$

Both p and $h^{-1}q_jhp$ belong to L , so $h^{-1}q_jh \in \mathcal{F}$. Only finitely many ordered tuples

$$(h^{-1}q_1h, \dots, h^{-1}q_mh)$$

are therefore possible. Choose one representative h_i for each tuple that occurs, with $1 \leq i \leq N$.

If h has the same tuple as h_i , then, for every j ,

$$h^{-1}q_jh = h_i^{-1}q_jh_i \implies (hh_i^{-1})q_j(hh_i^{-1})^{-1} = q_j.$$

Consequently $hh_i^{-1} \in C$, or equivalently

$$h \in Ch_i, \quad h = ch_i \quad \text{for some } c \in C.$$

Here Ch_i is a *right* coset, in the standard convention. Now set

$$K_C = Y_A \cap \bigcup_{i=1}^N h_i K.$$

This set is compact. Given $x \in Y_A$, write $x = hp$ with $p \in K$, choose its tuple representative h_i , and write $h = ch_i$ as above. Then $c^{-1}x = h_i p$ belongs to both Y_A and $h_i K$, so $x \in CK_C$. Thus $CK_C = Y_A$, proving cocompactness.

Finally, for $y \in Y_A$, $c \in C$, and every positive integer n , invariance and the inherited metric give

$$d_{Y_A}(y, c^n y) = d_X(y, c^n y).$$

Dividing by n and taking limits proves equality of stable lengths. $\square$

Proof of Theorem 1.1. Let M be the explicit finite matrix of Definition 7.1. Suppose, for a contradiction, that $G = A_M$ acts properly and cocompactly by isometries on a nonempty proper CAT(0) space X . No restriction is placed on the dimension of X .

Choose $x_0 \in X$ and a finite number

$$A \geq \max\{d_X(x_0, dx_0), d_X(x_0, ex_0)\}.$$

The set

$$Y = \{x \in X : d_X(x, dx) \leq A, d_X(x, ex) \leq A\}$$

contains x_0 . By Lemmas 7.4 and 7.5, T acts geometrically on Y , and its stable lengths there agree with its stable lengths in X .

Sending every Artin generator of G to $1 \in \mathbb{Z}$ defines a homomorphism $G \rightarrow \mathbb{Z}$: both sides of each defining Artin relation have the same number of letters. In particular, every squared generator maps to 2 and has infinite order. Lemma 2.2 therefore gives $\ell_X((s_1^{uv})^2) > 0$ for each $(u, v) \in \mathcal{E}$. The corresponding embedded B_{11} block satisfies all the relations required by Proposition 3.1, so that

$$\frac{\ell_X((uv)^3)}{3\ell_X(u^2)} = \frac{\ell_X((uv)^3)}{3\ell_X(v^2)} > \frac{1}{2} \quad ((u, v) \in \mathcal{E}).$$

Equality of stable lengths transfers all three strict inequalities to the geometric T -action on Y . This contradicts Proposition 4.1.

The 29-generator Artin group A_M therefore has no proper cocompact isometric action on any nonempty proper CAT(0) space. This is the required counterexample. $\square$

References

- [1] N. Brady and J. Crisp, *Two-dimensional Artin groups with $CAT(0)$ dimension three*, Geom. Dedicata **94** (2002), 185–214. doi:[10.1023/A:1020962804856](https://doi.org/10.1023/A:1020962804856). Preprint: [arXiv:math/0012102v2](https://arxiv.org/abs/math/0012102v2).
- [2] T. Brady and J. P. McCammond, *Three-generator Artin groups of large type are biautomatic*, J. Pure Appl. Algebra **151** (2000), no. 1, 1–9. doi:[10.1016/S0022-4049\(99\)00094-8](https://doi.org/10.1016/S0022-4049(99)00094-8). Preprint: [arXiv:math/9809174v1](https://arxiv.org/abs/math/9809174v1).
- [3] T. Brady and J. McCammond, *Braids, posets and orthoschemes*, Algebr. Geom. Topol. **10** (2010), no. 4, 2277–2314. doi:[10.2140/agt.2010.10.2277](https://doi.org/10.2140/agt.2010.10.2277).
- [4] M. R. Bridson, *Length functions, curvature and the dimension of discrete groups*, Math. Res. Lett. **8** (2001), 557–567. [Author’s preprint page](#).
- [5] M. R. Bridson and A. Haefliger, *Metric Spaces of Non-Positive Curvature*, Grundlehren der mathematischen Wissenschaften, vol. 319, Springer-Verlag, Berlin, 1999. doi:[10.1007/978-3-662-12494-9](https://doi.org/10.1007/978-3-662-12494-9).
- [6] R. Charney, *Problems related to Artin groups*, American Institute of Mathematics, undated problem list. [Author’s manuscript](#).
- [7] R. Charney and M. W. Davis, *Finite $K(\pi, 1)$ ’s for Artin groups*, in *Prospects in Topology*, F. Quinn (ed.), Annals of Mathematics Studies, vol. 138, Princeton University Press, Princeton, NJ, 1995, pp. 110–124. [Author’s preprint](#).
- [8] J. Crisp, *On the $CAT(0)$ dimension of 2-dimensional Bestvina–Brady groups*, Algebr. Geom. Topol. **2** (2002), 921–936. doi:[10.2140/agt.2002.2.921](https://doi.org/10.2140/agt.2002.2.921). [arXiv:math/0211130](https://arxiv.org/abs/math/0211130).
- [9] J. Crisp and L. Paoluzzi, *On the classification of $CAT(0)$ structures for the 4-string braid group*, Michigan Math. J. **53** (2005), no. 1, 133–163. doi:[10.1307/mmj/1114021090](https://doi.org/10.1307/mmj/1114021090).
- [10] F. Digne, *Présentation des groupes de tresses purs et de certaines de leurs extensions*, unpublished manuscript, 1999; [arXiv version 3](https://arxiv.org/abs/1511.08731v3), 2016. [arXiv:1511.08731v3](https://arxiv.org/abs/1511.08731v3).
- [11] S. M. Gersten, *Reducible diagrams and equations over groups*, in *Essays in Group Theory*, Mathematical Sciences Research Institute Publications, vol. 8, Springer-Verlag, New York, 1987, pp. 15–73. doi:[10.1007/978-1-4613-9586-7_2](https://doi.org/10.1007/978-1-4613-9586-7_2).
- [12] T. Haettel, *XXL type Artin groups are $CAT(0)$ and acylindrically hyperbolic*, Ann. Inst. Fourier (Grenoble) **72** (2022), no. 6, 2541–2555. doi:[10.5802/aif.3524](https://doi.org/10.5802/aif.3524).
- [13] T. Haettel, D. Kielak, and P. Schwer, *The 6-strand braid group is $CAT(0)$* , Geom. Dedicata **182** (2016), 263–286. doi:[10.1007/s10711-015-0138-9](https://doi.org/10.1007/s10711-015-0138-9). Preprint: [arXiv:1304.5990v2](https://arxiv.org/abs/1304.5990v2).
- [14] S. G. Jeong, *The seven-strand braid group is $CAT(0)$* , Manuscripta Math. **171** (2023), 563–581. doi:[10.1007/s00229-022-01387-x](https://doi.org/10.1007/s00229-022-01387-x). Preprint: [arXiv:2009.09350v1](https://arxiv.org/abs/2009.09350v1).
- [15] M. Rodenhausen, *Centralisers of polynomially growing automorphisms of free groups*, Ph.D. thesis, Rheinische Friedrich-Wilhelms-Universität Bonn, 2013. [hdl:20.500.11811/5725](https://nbn-resolving.org/urn:nbn:de:hbz:5:1-5725).
- [16] K. E. Ruane, *Dynamics of the action of a $CAT(0)$ group on the boundary*, Geom. Dedicata **84** (2001), 81–99. doi:[10.1023/A:1010301824765](https://doi.org/10.1023/A:1010301824765).

- [17] J.-P. Serre, *Trees*, translated from the French by J. Stillwell, Springer-Verlag, Berlin–New York, 1980. doi:[10.1007/978-3-642-61856-7](https://doi.org/10.1007/978-3-642-61856-7).
- [18] E. L. Swenson, *A cut point theorem for $CAT(0)$ groups*, J. Differential Geom. **53** (1999), no. 2, 327–358. doi:[10.4310/jdg/1214425538](https://doi.org/10.4310/jdg/1214425538).